Vaibhav Mahore

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B.Tech in Mathematics and Computing GitHub LeetCode

EDUCATION

Indian Institute of Science (IISc), Bangalore *Aug 2023 - Present*
Bachelor of Technology (B.Tech) in Mathematics and Computing
Relevant Courses: Data Structures and Algorithms, Linear Algebra, Probability, Machine Learning, Deep Learning

St. Ann's Senior Secondary School, CBSE *Class XII: 2023, Class X: 2021*

EXPERIENCE

AI/ML Research Intern, Ericsson India Private Limited *Bangalore, May 2026 - July 2026*

- Benchmarked WiMamba vs. Transformer on AWS A10G GPUs, proving $\mathcal{O}(\mathbf{T})$ linear scaling with **10.7× lower latency** (30ms vs 327ms) and **42.6× less GPU memory** (113MB vs 4.8GB) at 4×4 patch size.
- Evaluated geographic OOD transfer (Florida→San Diego) across 5 tasks, achieving a **0.9626 composite score** (+16.6% over baseline) and **7.4-7.7 dB lower NMSE** on channel estimation and interpolation.
- Explored a Soft Unfreezing Parameter-Efficient Fine-Tuning strategy using PyTorch linear probing, freezing 75% of parameters while retaining **98.5% of full fine-tuning performance**.

Machine Learning Engineer (AI Data Trainer), Alignerr *Remote, May 2025 - Aug 2025*
Authored structured reasoning traces and high-quality training data to improve LLM decision-making, tool selection, and step-by-step problem-solving across complex technical tasks.

Machine Learning Engineer, micro1 *Remote, May 2025 - Aug 2025*

- Contributed high-quality data annotation, evaluation, and model validation workflows for frontier AI systems.
- Performed data preprocessing and supported machine learning model development using TensorFlow and Scikit-learn.

PROJECTS

Efficient Sequence Modeling: From S4 to Mamba-3 *Sep 2026*

- Implemented minimal S4D, Mamba, Mamba-2 and Mamba-3 in pure PyTorch from first principles, including a chunked State-Space-Duality path test-verified against a sequential scan; benchmarked latency, memory and throughput at 1K–32K sequence lengths on a 6 GB RTX 4050.
- Reproduced the Mamba-3 paper's capability claim: complex-valued transitions solve parity tracking (**98.3%**) where real-transition SSMs and Transformers stay at chance (≤56%); on selective copying, LTI S4D collapses (**6%**) vs **67%** for selective SSMs.

Tech Stack: Python, PyTorch, NumPy, SciPy, Hugging Face, Matplotlib, pytest

AIRE: AI Reliability & Evaluation Engine *Sep 2026*

- Built a trace-based evaluation engine for LLM/RAG systems: replayable run storage, deterministic metrics (groundedness, citation coverage, retrieval recall/MRR), failure taxonomy, and a direction-aware regression engine with per-case attribution.
- Evaluated three RAG versions over a 26-case suite, flagging an intentional fluency-for-grounding regression (groundedness **0.57→0.00**); a live Gemini responder then fixed abstention (0.0→**1.0**) and grounding (**1.0**) at **15 s**/case.

Tech Stack: Python, pytest, HTML/CSS

Autonomous Repository Engineer *Sep 2026*

- Built a verified tool-using coding agent: a deterministic state machine (PLAN→ACT→ OBSERVE→VERIFY) wraps the planner, which only proposes actions; seven validated tools with schema checks, a workspace path jail, AUTO/ASK/DENY permission policies and command allowlists.
- Benchmark: **4/4** SWE tasks deterministically with **zero failed tool calls**; a live Gemini planner passes **2/2** on the same runtime. Completion requires green test evidence, never model claims.

Tech Stack: Python, pytest, Git

SKILLS

- **Languages:** Python, C++, Basic SQL
- **AI/ML:** PyTorch, TensorFlow, Scikit-learn, NumPy, SciPy, Pandas, Hugging Face Datasets, Matplotlib, pytest
- **LLM & AI Systems:** LLM evaluation & observability (tracing, judges, regression testing), RAG, tool-using agents, sequence modeling (Transformers, State Space Models)
- **MLOps & Cloud:** Docker, AWS (EC2, S3), Linux, Git/GitHub
- **Core Concepts:** Deep Learning, Model Architecture, Experiment Design & Reproducibility, Software Engineering Practice (testing, sandboxing, CI-style verification)

CERTIFICATIONS

Oracle Certified: Generative AI Professional Credential & Data Science Professional Credential *Aug 2025*

ACHIEVEMENTS & POSITIONS OF RESPONSIBILITY

Sci-Tech Coordinator, Rhapsody Fest, IISc *2023 - 2026*

- Led end-to-end organization of three inter-college gaming tournaments and a competitive coding contest, managing the full event lifecycle for **200+ registered teams**.
- Administered a **500+ member** Discord community for registration and tech support; drove fest publicity through digital campaigns across colleges.
- **All India Rank 2412 in JEE Advanced 2023** (top 1.34% among 180,000+ candidates nationwide).
- Organized and conducted a **Competitive Coding Contest** at Rhapsody Fest, IISc as Sci-Tech Coordinator, overseeing problem curation and execution for **200+ participants** across multiple colleges.